

WANGARATTA, Australia

WMO: 948890

Lat: 36.4206S

Lon: 146.3056E

Elev: 153

StdP: 99.50

Time Zone: 10.00 (AUS)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-1.9	-0.8	-5.1	2.5	22.5	-3.2	3.0	14.7	9.8	10.2	8.8	10.5	0.8	210	0.486

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.1	37.3	18.2	35.0	17.9	33.0	17.5	21.5	27.5	20.5	27.3	19.7	27.6	4.9	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.1	15.1	22.7	18.8	13.9	22.1	17.2	12.5	21.3	63.2	27.0	59.7	27.1	56.8	27.6	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.2	8.1	7.1	-3.6	41.2	1.2	2.0	-4.4	42.7	-5.1	43.9	-5.8	45.0	-6.7	46.5		
			-3.7	23.0	1.2	1.6	-4.6	24.1	-5.3	25.0	-6.0	25.9	-6.9	27.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	14.7	23.1	22.3	19.2	14.5	10.7	8.3	7.8	8.5	10.8	13.7	17.6	20.5	(6)
(7)		DBStd	6.16	3.71	3.46	3.35	2.87	2.58	2.24	2.07	2.08	2.41	2.99	3.54	3.63	(7)
(8)		HDD10.0	234	0	0	0	2	23	61	72	58	17	3	0	0	(8)
(9)		HDD18.3	1778	6	6	30	118	238	302	326	306	225	148	56	18	(9)
(10)		CDD10.0	1951	407	344	284	138	44	9	6	9	42	116	228	324	(10)
(11)		CDD18.3	452	154	117	56	4	0	0	0	0	3	33	84	(11)	
(12)		CDH23.3	7026	2261	1596	856	125	2	0	0	0	8	154	655	1370	(12)
(13)		CDH26.7	3346	1240	791	338	19	0	0	0	1	42	265	652	(13)	
(14)	Wind	WSAvg	2.8	3.4	3.0	2.8	2.2	2.1	2.1	2.4	2.6	2.9	3.2	3.5	(14)	
(15)	Precipitation	PrecAvg	653	43	53	37	39	54	69	70	64	62	54	57	52	(15)
(16)		PrecMax	981	137	189	109	102	114	140	134	119	167	152	122	132	(16)
(17)		PrecMin	296	0	0	1	1	0	31	16	8	14	0	14	4	(17)
(18)		PrecStd	167	37	54	28	29	35	30	31	32	35	40	30	36	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.1	39.2	35.8	29.0	22.2	17.9	16.2	19.6	24.2	31.0	35.5	38.8	(19)
(20)			MCWB	18.6	18.9	17.9	16.3	14.1	12.3	11.4	13.4	15.0	15.8	17.1	17.5	(20)
(21)		2%	DB	38.1	36.0	32.8	26.2	20.1	16.0	14.9	17.0	21.2	27.2	32.2	35.2	(21)
(22)			MCWB	18.4	18.4	17.5	15.4	13.5	11.5	10.9	11.8	14.2	15.3	16.4	17.2	(22)
(23)		5%	DB	35.2	33.8	30.2	24.6	18.5	14.8	13.6	15.2	19.2	24.8	29.8	32.8	(23)
(24)			MCWB	17.9	18.1	16.9	15.0	13.0	11.0	10.2	10.7	13.1	15.0	16.3	16.5	(24)
(25)		10%	DB	33.1	31.7	28.1	22.7	16.8	13.3	12.2	13.9	17.6	22.1	27.1	30.2	(25)
(26)			MCWB	17.8	17.6	16.3	14.3	12.4	10.5	9.4	10.2	12.3	14.3	15.8	16.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.8	23.1	21.3	18.7	16.4	14.2	13.1	14.1	16.7	19.0	21.0	21.6	(27)
(28)			MCDWB	27.5	28.5	26.2	24.2	18.3	15.3	14.4	17.3	20.8	24.9	27.4	24.9	(28)
(29)		2%	WB	21.3	21.6	19.8	17.2	14.9	12.8	11.6	12.7	15.3	17.4	19.5	20.1	(29)
(30)			MCDWB	28.4	27.7	25.7	21.6	17.8	14.5	13.6	15.8	19.8	22.4	25.9	26.2	(30)
(31)		5%	WB	20.4	20.5	18.7	16.1	13.9	11.7	10.8	11.4	14.0	16.2	18.3	18.9	(31)
(32)			MCDWB	28.6	27.4	26.4	21.7	17.1	13.7	12.7	14.5	18.1	21.5	24.7	26.4	(32)
(33)		10%	WB	19.4	19.4	17.8	15.2	12.9	10.8	10.0	10.5	12.8	15.2	17.2	17.9	(33)
(34)			MCDWB	29.2	27.9	25.8	21.1	16.0	12.9	11.7	13.2	16.5	20.9	23.6	27.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	17.1	16.3	15.9	15.2	12.5	10.7	9.7	11.0	12.7	14.9	15.9	16.9	(35)
(36)			MCDBR	21.2	19.7	19.2	18.3	15.3	13.5	12.0	14.2	16.5	20.6	21.0	21.4	(36)
(37)			MCWBR	5.9	6.0	7.0	8.4	8.9	9.2	8.2	9.3	9.7	9.9	7.3	6.5	(37)
(38)		5% WB	MCDBR	15.1	14.1	14.8	14.6	11.5	10.4	9.4	12.2	14.0	15.4	14.7	15.3	(38)
(39)			MCWBR	5.6	5.4	6.3	7.3	7.3	7.6	6.9	8.3	8.8	8.7	6.6	6.2	(39)
(40)																
(41)	Clear-Sky Solar Irradiance	taub		0.356	0.349	0.322	0.315	0.302	0.294	0.290	0.298	0.331	0.331	0.346	0.349	(41)
(42)		taud		2.455	2.496	2.569	2.551	2.561	2.584	2.595	2.553	2.445	2.475	2.472	2.469	(42)
(43)		Ebn at Noon		978	962	952	899	853	834	861	904	925	970	982	989	(43)
(44)		Edh at Noon		118	109	94	84	72	66	69	82	104	110	116	118	(44)
(45)	All-Sky Solar Radiation	RadAvg		7.67	6.71	5.55	3.96	2.69	2.04	2.19	3.09	4.44	5.93	6.91	7.66	(45)
(46)		RadStd		0.44	0.41	0.28	0.21	0.20	0.26	0.15	0.27	0.44	0.52	0.37	0.34	(46)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.45	N/A	N/A	+1.59	N/A	N/A	N/A	N/A	N/A	+109	(46)
(47)	Regional (35 neighbors)	+0.45	N/A	N/A	+0.97	N/A	N/A	N/A	-97	+153	+70	(47)

Nomenclature: See separate page